Distribution-Free Uncertainty Quantification in Machine Learning: A Theoretical and Empirical Comparison of Conformal Prediction, Bayesian Credible Intervals, and Parametric Bootstrap in R

Yuvraj Chandak
School of Computer Science and Engineering
Register No.: 25BCE2561
Vellore Institute of Technology, Vellore, Tamil Nadu, India
Email: yuvraj.chandak2025@vitstudent.ac.in

Dr. Veermani S
Department of Mathematics
Project Supervisor & Professor
Vellore Institute of Technology, Vellore, Tamil Nadu, India
Email: veermani.s@vit.ac.in

***Abstract*—Standard machine learning models typically output single point predictions. While these point forecasts can be accurate on average, they provide no information about predictive uncertainty, dispersion, or tail risk. Classical uncertainty quantification methods —such as Ordinary Least Squares (OLS) prediction intervals, the parametric bootstrap, and Bayesian Normal-Inverse-Gamma conjugate models—rely on strong distributional assumptions, including Gaussian errors, constant variance (homoscedasticity), and correct likelihood specification. In contrast, conformal prediction is a distribution-free framework that guarantees valid prediction interval coverage in finite samples, assuming only that the data points are exchangeable. In this paper, we present a systematic theoretical and empirical comparison of five uncertainty quantification methods in R: parametric OLS, the parametric bootstrap, Bayesian Normal-Inverse-Gamma inference, Split Conformal Prediction, and Conformalized Quantile Regression (CQR). We evaluate these methods across $M = 100$ Monte Carlo replications under three distinct error regimes: standard Gaussian noise, heavy-tailed Student-t_3 noise, and heteroscedastic skewed noise, as well as on the benchmark Boston Housing dataset. Our results show that when Gaussian and homoscedastic assumptions fail, classical parametric and Bayesian methods suffer from coverage inflation (92.7% to 93.0% at a nominal 90% level) because extreme residuals inflate the estimated error variance. Conformal methods, by contrast, maintain empirical coverage consistent with the nominal 90% target across all tested regimes. Furthermore, while standard split conformal intervals have fixed width, CQR dynamically adapts its interval width to local noise levels, achieving significantly lower Winkler scores under heteroscedasticity (9.756 ± 0.897 versus 10.096 ± 0.856 for OLS; Wilcoxon $p = 2.22 \times 10^{-10}$). We also analyze practical trade-offs regarding calibration sample sizes, training data costs, and evaluate performance on real-world housing data. All code, datasets, and experiment scripts are made publicly available.**

***Index Terms*—Conformal Prediction, Exchangeability, Bayesian Inference, Parametric Bootstrap, Uncertainty Quantification, Finite-Sample Coverage, Non-Parametric Statistics, Machine Learning in R.**

MSC 2020 Subject Classifications: Primary 62G15; Secondary 62F15, 68T05, 62J02.

I. Introduction & Problem Formulation

Supervised machine learning algorithms, such as Random Forests [3], Gradient Boosted Decision Trees [5], and deep neural networks [8], are widely used across science and industry to make predictions from complex, high-dimensional data. Given a training dataset $D_n = \{(X_i, Y_i)\}_{i=1}^n$ drawn from an unknown joint probability distribution $P_{X,Y}$ on $X \times R$, where $X \subseteq R^d$, model training typically minimizes an empirical loss function:

$$\hat{R}_n(f) = \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}(Y_i, f(X_i)),$$

While empirical risk minimization can produce models with excellent average accuracy, point predictions $\hat{Y}_{n+1} = \hat{f}_n(X_{n+1})$ do not tell the user how confident the model is in any specific forecast. In high-stakes applications like medical diagnostics, financial risk management, and autonomous systems, decisions require reliable measures of uncertainty. Uncertainty quantification (UQ) addresses this need by producing a prediction interval $\hat{C}_n(X_{n+1}) = [\hat{L}(X_{n+1}), \hat{U}(X_{n+1})] \subset R$ that covers the true future response Y_{n+1} with a user-specified probability $1 - \alpha \in (0, 1)$:

$$\mathbb{P}(Y_{n+1} \in \hat{C}_n(X_{n+1})) \geq 1 - \alpha.$$

A. Aleatoric and Epistemic Uncertainty Decomposition

Predictive uncertainty can be broken down into two distinct sources:

$$\mathrm{Var}(Y \mid X = x) = \sigma^2(x) + \mathrm{Var}_{\mathcal{D}_n}(\hat{f}_n(x)),$$

1) *Aleatoric Uncertainty:* This represents the inherent randomness in the data-generating process (such as measurement noise or natural variability). It cannot be eliminated simply by collecting more data. When the error variance $\mathrm{Var}(\varepsilon \mid X = x) = \sigma^2$ is constant across all inputs x, the data are homoscedastic. When the error variance varies with x, the data are heteroscedastic.

2) *Epistemic Uncertainty:* This reflects the model's lack of knowledge due to limited training data, parameter estimation uncertainty, and model misspecification. In parametric models, epistemic uncertainty typically decreases at rate $O_p(n^{-1/2})$ as sample size increases.

Classical statistical frameworks—including Ordinary Least Squares (OLS) prediction intervals, parametric bootstrap resamples, and Bayesian credible intervals—estimate both uncertainty components under the assumption that the data follow a specified parametric family (typically a Gaussian distribution with constant variance). When these assumptions are violated by heavy tails,

skewness, or heteroscedasticity, classical methods can produce miscalibrated prediction intervals.

B. Scope and Objectives

Conformal prediction offers an appealing alternative: it guarantees valid marginal coverage in finite samples without requiring parametric distributional assumptions, relying only on data exchangeability (a property satisfied by independent and identically distributed data) [1], [19].

While the theoretical foundations of split conformal prediction [14], [12] and Conformalized Quantile Regression [15] are established, practical comparisons across paradigms under identical data splits and varying noise conditions remain valuable for practitioners. Furthermore, while modern R packages like `brms` [4], `tidymodels` [10], and `probably` [11] implement various facets of UQ, a unified comparison across these exact regimes has been lacking. The goals of this paper are:

1) *Unified Comparative Study in R:* We evaluate five distinct uncertainty quantification methods—parametric OLS, the parametric bootstrap, Bayesian Normal-Inverse-Gamma inference, Split Conformal Random Forests, and Conformalized Quantile Regression (CQR)—using a consistent evaluation framework in R.

2) *Stress-Testing Under Misspecification:* We test how robust each method is when its underlying assumptions fail, using synthetic datasets with heavy-tailed Student-t_3 noise ($\kappa = \infty$) and heteroscedastic, right-skewed errors ($\sigma(X_1) = 1 + 2|X_1|$).

3) *Formal Statistical Evaluation:* We measure performance using replication-level Student's *t*-tests for empirical coverage, interval width, and strictly proper Winkler scores (S_α) to assess overall interval quality.

4) *Analysis of Practical Trade-offs:* We investigate the cost of reserving calibration data, examine the asymptotic behavior of fixed-variance bootstrap intervals, and study performance on the real-world Boston Housing benchmark dataset.

The remainder of this paper is structured as follows. Section II reviews mathematical foundations and derivations for all five methods, including a complete proof of the finite-sample conformal coverage theorem. Section III describes experimental design, noise regimes, and evaluation metrics. Section IV presents empirical findings, statistical tests, and graphical analyses. Section V discusses theoretical considerations, including sample-splitting trade-offs, conditional coverage limitations, and covariate shift. Section VI concludes the paper.

II. Mathematical Foundations

A. Classical Ordinary Least Squares Prediction Intervals

The classical linear regression model assumes $Y = X\beta + \varepsilon$, where $\varepsilon \sim N_n(0, \sigma^2 I_n)$, $X \in \mathbb{R}^{n \times p}$ is a design matrix of full column rank ($p < n$), and $\beta \in \mathbb{R}^p$ is unknown. The OLS estimator is $\hat{\beta} = (X^\top X)^{-1} X^\top Y$ with sampling distribution $\hat{\beta} \sim N_p(\beta, \sigma^2 (X^\top X)^{-1})$. The unbiased estimator of σ^2 is residual mean squared error $\hat{\sigma}^2 = \|Y - X\hat{\beta}\|_2^2 / (n - p)$. By Cochran's Theorem, $(n - p)\hat{\sigma}^2 / \sigma^2 \sim \chi^2_{n-p}$ independently of $\hat{\beta}$.

For a new test point (X_{n+1}, Y_{n+1}) where $X_{n+1} = x_{n+1} \in \mathbb{R}^p$ and $Y_{n+1} = x_{n+1}^\top \beta + \varepsilon_{n+1}$ with $\varepsilon_{n+1} \sim N(0, \sigma^2)$ independent of training data D_n, the point prediction is $\hat{Y}_{n+1} = x_{n+1}^\top \hat{\beta}$. The prediction error is $e_{n+1} = Y_{n+1} - \hat{Y}_{n+1} = \varepsilon_{n+1} - x_{n+1}^\top (\hat{\beta} - \beta)$. Because ε_{n+1} and $\hat{\beta}$ are independent Gaussian variables, the conditional mean is $E[e_{n+1} \mid X, x_{n+1}] = 0$ and variance:

$$\mathbf{Var}(e_{n+1} \mid \boldsymbol{X}, \boldsymbol{x}_{n+1}) = \sigma^2 \left(1 + \boldsymbol{x}_{n+1}^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_{n+1}\right).$$

Defining query leverage $h_{n+1} = x_{n+1}^\top (X^\top X)^{-1} x_{n+1}$ and substituting $\hat{\sigma}$ for σ, the studentized prediction pivot satisfies $T = (Y_{n+1} - x_{n+1}^\top \hat{\beta}) / [\hat{\sigma} \sqrt{(1 + h_{n+1})}] \sim t_{n-p}$. This gives the exact $(1 - \alpha)$ parametric prediction interval:

$$\hat{C}_{\text{param}}(\boldsymbol{x}_{n+1}) = \left[\boldsymbol{x}_{n+1}^\top \hat{\boldsymbol{\beta}} \pm t_{1-\alpha/2,\, n-p}\, \hat{\sigma} \sqrt{1 + h_{n+1}}\right].$$

B. Parametric Bootstrap Prediction Intervals

The parametric bootstrap generates prediction intervals by simulating synthetic outcomes from the fitted parametric distribution rather than relying on theoretical pivot distributions: 1) Fit the base model to obtain $\hat{\beta}$ and residual variance $\hat{\sigma}^2$. 2) For $b = 1, \ldots, B$, draw synthetic training residuals $\varepsilon^{*(b)} \sim N_n(0, \hat{\sigma}^2 I_n)$, form bootstrap responses $Y^{*(b)} = X\hat{\beta} + \varepsilon^{*(b)}$, re-estimate coefficients $\hat{\beta}^{*(b)} = (X^\top X)^{-1} X^\top Y^{*(b)}$, draw a synthetic test error $\varepsilon_{\text{test}}^{*(b)} \sim N(0, \hat{\sigma}^2)$, and compute the simulated test prediction:

$$Y_{\text{test}}^{*(b)} = \boldsymbol{x}_{\text{test}}^\top \hat{\boldsymbol{\beta}}^{*(b)} + \epsilon_{\text{test}}^{*(b)}.$$

3) The empirical $(1 - \alpha)$ bootstrap interval is formed from the percentiles: $\hat{C}_{\text{boot}}(x_{\text{test}}) = [\hat{q}_{\alpha/2}(\{Y_{\text{test}}^{*(b)}\}_{b=1}^{B}),\ \hat{q}_{1-\alpha/2}(\{Y_{\text{test}}^{*(b)}\}_{b=1}^{B})]$.

Asymptotic Behavior of the Fixed-$\hat{\sigma}$ Bootstrap: Because $\hat{\sigma}$ is held fixed across resamples rather than re-estimated or drawn from a posterior, conditional on $(\hat{\beta}, \hat{\sigma})$ the simulated test response satisfies $Y_{\text{test}}^{*(b)} \sim N(x_{\text{test}}^\top \hat{\beta}, \hat{\sigma}^2(1 + h_{\text{test}}))$. As $B \to \infty$, empirical quantiles converge almost surely to normal quantiles $[x_{\text{test}}^\top \hat{\beta} \pm z_{1-\alpha/2}\, \hat{\sigma}\, \sqrt{(1 + h_{\text{test}})}]$. Because $z_{1-\alpha/2} < t_{1-\alpha/2,\, n-p}$, the bootstrap interval is slightly narrower than the exact OLS interval, leading to modest under-coverage in finite samples under Gaussian errors (89.21% in our experiments).

C. Bayesian Credible Intervals & Misspecification

In Bayesian regression, predictive uncertainty is quantified through the posterior predictive distribution. Under standard uninformative Jeffreys prior limits on design matrix X, the marginal posterior predictive distribution is Student-*t* with degrees of freedom $\nu = n - p$, location $x_{n+1}^\top \hat{\beta}$, and scale $s\sqrt{(1 + h_{n+1})}$:

$$\boldsymbol{y}_{n+1} \mid \boldsymbol{x}_{n+1}, \mathcal{D}_n \sim t_{n-p}\left(\boldsymbol{x}_{n+1}^\top \hat{\boldsymbol{\beta}},\ \boldsymbol{s}^2(1 + \boldsymbol{h}_{n+1})\right),$$

yielding equal-tailed credible intervals numerically identical to OLS: $\hat{C}_{\text{Bayes}}(x_{n+1}) = [x_{n+1}^\top \hat{\beta} \pm t_{1-\alpha/2,\, n-p}\, s\, \sqrt{(1 + h_{n+1})}]$.

Misspecification and the Bernstein–von Mises Theorem: Under a correctly specified likelihood, the Bernstein–von Mises Theorem (van der Vaart [17]) guarantees that the posterior distribution of $\sqrt{n}(\theta - \theta_0)$ converges to a Gaussian distribution with covariance

equal to the inverse Fisher information. However, when the true distribution P_0 is not in the assumed model family, the posterior concentrates around pseudo-true parameter θ^* with variance $\Sigma_{\text{Bayes}} = H(\theta^*)^{-1}$, diverging from the true frequentist sandwich covariance $\Sigma_{\text{freq}} = H(\theta^*)^{-1}J(\theta^*)H(\theta^*)^{-1}$ (Kleijn and van der Vaart [9]). Because $H \neq J$ under misspecification, Bayesian credible intervals do not guarantee valid frequentist coverage when the likelihood model is incorrect.

D. Finite-Sample Conformal Prediction

Conformal prediction (Vovk et al. [19]) provides valid predictive inference without requiring parametric assumptions about the data distribution:

Definition 1 (Exchangeability): *Random variables $Z_1, ..., Z_{n+1}$ are exchangeable if for every permutation π of indices $\{1, ..., n+1\}$, $(Z_1, ..., Z_{n+1}) \sim^d (Z_{\pi(1)}, ..., Z_{\pi(n+1)})$.*

In Split Conformal Prediction (Papadopoulos et al. [14]; Lei et al. [12]), training data D_n is partitioned into a proper training set D_1 (size n_1) and calibration set D_2 (size n_2). A regression model $\hat{\mu}$ is fitted on D_1. Non-conformity residuals are computed on calibration points: $R_i = |Y_i - \hat{\mu}(X_i)|$, $\forall i \in D_2$. Sorting calibration scores in ascending order $R_{(1)} \leq ... \leq R_{(n_2)}$, the conformal threshold is $\hat{q}_{1-\alpha} = R_{(k)}$ with index $k = \lceil (n_2 + 1)(1 - \alpha) \rceil$ (with $\hat{q} = \infty$ if $k > n_2$). The split conformal prediction interval is $\hat{C}_n(X_{n+1}) = [\hat{\mu}(X_{n+1}) \pm \hat{q}_{1-\alpha}]$.

Theorem 1 (Finite-Sample Marginal Coverage): *Let $Z_1, ..., Z_{n+1}$ be exchangeable. Let $\hat{\mu}$ be trained on D_1, and calibration scores R_i computed on D_2. Then for any $\alpha \in (0, 1)$, setting $k = \lceil (n_2 + 1)(1 - \alpha) \rceil$ unconditionally guarantees:*

$$\mathbb{P}\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \geq 1 - \alpha.$$

Furthermore, if non-conformity scores have almost surely distinct values (e.g., continuous distributions):

$$\mathbb{P}\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \leq 1 - \alpha + \frac{1}{n_2 + 1}.$$

Proof: Condition on training set D_1, treating $\hat{\mu}$ as fixed. Because $Z_1, ..., Z_{n+1}$ are exchangeable and $\hat{\mu}$ depends only on D_1, the remaining $n_2 + 1$ residuals $(R_1, ..., R_{n_2}, R_{n+1})$ are conditionally exchangeable. By symmetry, the test score rank satisfies $\mathrm{P}(\mathrm{rank}(R_{n+1}) = j \mid D_1) = 1 / (n_2 + 1)$ for every $j \in \{1, ..., n_2 + 1\}$. Coverage occurs if and only if $R_{n+1} \leq R_{(k)} \Leftrightarrow \mathrm{rank}(R_{n+1}) \leq k$. Summing over ranks yields $\mathrm{P}(Y_{n+1} \in \hat{C}_n \mid D_1) = k / (n_2 + 1)$. By definition of the ceiling function, $(n_2 + 1)(1 - \alpha) \leq k < (n_2 + 1)(1 - \alpha) + 1$, which implies $1 - \alpha \leq k / (n_2 + 1) < 1 - \alpha + 1 / (n_2 + 1)$. Taking the expectation over D_1 completes the proof. Ties without randomization can only increase inclusion probability, preserving the lower bound. ∎

E. Conformalized Quantile Regression (CQR)

While Split Conformal Prediction guarantees marginal coverage, its interval width $2\hat{q}_{1-\alpha}$ is constant across X. Under heteroscedastic noise $\mathrm{Var}(\varepsilon \mid X = x) = \sigma^2(x)$, constant-width bands fail to adapt: they are unnecessarily wide where $\sigma^2(x)$ is small and too narrow where $\sigma^2(x)$ is large.

To address this, Romano, Patterson, and Candès [15] introduced Conformalized Quantile Regression (CQR): train lower and upper conditional quantile models on D_1 via Quantile Random Forests (Meinshausen [13]) by minimizing the pinball loss. On calibration points, compute signed conformity scores:

$$E_i = \max\{\hat{q}_{\alpha/2}(X_i) - Y_i,\ Y_i - \hat{q}_{1-\alpha/2}(X_i)\}.$$

The conformal cutoff is $Q_{1-\alpha} = E_{(k)}$ with index $k = \lceil (n_2 + 1)(1 - \alpha) \rceil$. The adjusted CQR interval is:

$$\hat{C}_{\text{CQR}}(X_{n+1}) = \big[\hat{q}_{\alpha/2}(X_{n+1}) - Q_{1-\alpha},\ \hat{q}_{1-\alpha/2}(X_{n+1}) + Q_{1-\alpha}\big].$$

Theorem 1 of Romano et al. [15] establishes that CQR guarantees finite-sample marginal coverage $\mathrm{P}(Y_{n+1} \in \hat{C}_{\text{CQR}}) \geq 1 - \alpha$ under exchangeability alone. If base quantile estimators are consistent for the true conditional quantiles (Meinshausen [13]), $Q_{1-\alpha} \to^p 0$, asymptotically recovering conditional validity as well.

F. Cross-Conformal Prediction and CV+

Split conformal prediction incurs an efficiency loss because calibration data (n_2) are withheld from model training. To address this without the computational burden of full conformal prediction, Barber et al. [2] introduced CV+. Partition data into K folds, train model $\hat{\mu}_{-k}$ on remaining folds, and compute leave-fold-out residuals. Theorem 1 of Barber et al. [2] establishes that for symmetric base training algorithms, CV+ guarantees distribution-free coverage of $\mathrm{P}(Y_{n+1} \in \hat{C}_n^{\text{CV+}}) \geq 1 - 2\alpha - \min(2\alpha, 2/(n + 1))$, which simplifies to $1 - 2\alpha$. In empirical practice, CV+ coverage generally remains close to $1 - \alpha$.

III. Experimental Methodology

A. Simulation Design & Training Sample Budgets

We evaluate methods on a non-linear regression function $Y = 2\sin(\pi X_1) + X_2^2 - 1.5\,X_3 + \varepsilon$, with $X \sim \mathrm{U}[-1, 1]^3$ and total sample size $N = 1{,}500$ ($n_1 = 750$, $n_2 = 375$, $n_{\text{test}} = 375$), across $M = 100$ independent Monte Carlo replications under three noise regimes:

1) *Regime I (Homoscedastic Gaussian):* $\varepsilon \sim \mathrm{N}(0, 1^2)$. Standard homoscedastic baseline where Gaussian assumptions hold.

2) *Regime II (Heavy-Tailed Student-t):* $\varepsilon \sim t_{\nu=3}$. Heavy-tailed noise with infinite theoretical kurtosis ($\kappa = \infty$) and frequent large residuals.

3) *Regime III (Heteroscedastic Skewed):* $\varepsilon = (1 + 2|X_1|)\xi$, $\xi \sim \mathrm{Exp}(1) - 1$. The conditional standard deviation varies across the feature space from 1.0 to 3.0, and ξ has skewness $\gamma_1 = 2$.

Model Specifications & Training Allocation: The three parametric methods (OLS, Parametric Bootstrap, and Bayesian NIG) are fitted using polynomial basis expansion `y ~ poly(x1, 3) + poly(x2, 2) + x3` ($p = 7$ parameters) on the combined non-test sample $D_1 \cup D_2$ ($n_1 + n_2 = 1{,}125$ observations). Split Conformal Random Forest and CQR (`ranger`, 100 trees, `quantreg = TRUE`) train trees on $n_1 = 750$ observations using raw features (X_1, X_2, X_3) and hold out $n_2 = 375$ exclusively for

calibration, directly assessing the operational trade-off of conformal prediction.

B. Real-World Benchmark Dataset

We benchmark all methods on the `Boston` dataset from `MASS` ($N = 506$ observations, 13 features). Parametric models train on $D_1 \cup D_2$ (379 rows), while conformal models train on $n_1 = 253$ and calibrate on $n_2 = 126$ across $B = 50$ random splits (50% / 25% / 25% ratio; $n_{\text{test}} = 127$). The response variable is median home value `medv`.

C. Evaluation Metrics & Statistical Tests

Performance is quantified by: Empirical Coverage Rate (ECR); Mean Prediction Interval Width (MPIW); and strictly proper Winkler Score $S_\alpha = (\hat{U} - \hat{L}) + (2/\alpha)(\hat{L} - y)\mathrm{I}(y < \hat{L}) + (2/\alpha)(y - \hat{U})\mathrm{I}(y > \hat{U})$ with penalty slope 20 at nominal $\alpha = 0.10$ (Gneiting and Raftery [7]). Across $M = 100$ independent Monte Carlo replications, empirical coverage calibration is tested via a two-sided one-sample Student's t-test against nominal 90%:

$$t = \frac{\bar{p} - (1-\alpha)}{s_{\text{ECR}}/\sqrt{M}}, \quad \text{with } M - 1 = 99 \text{ degrees of freedom.}$$

For Boston housing, we report the sample mean and standard deviation across $B = 50$ splits. Because repeated splits share observations, metrics provide descriptive stability comparisons rather than independent hypothesis tests.

D. Reproducibility & Open-Source Repository

The complete, self-contained replication script, data preparation pipelines, and benchmark tables are hosted in an open-access repository: https://github.com/YuvrajC7/distribution-free-uncertainty-quantification. All simulations, statistical tests, and figures were executed in R version 4.6.1 under fixed random seed `set.seed(2026)`.

IV. Empirical Results & Discussion

This section presents empirical results obtained from running `simulation_experiments.R` with seed `2026`. Tables I–III summarize the Monte Carlo results across synthetic regimes, and Table IV reports the Boston Housing benchmark.

TABLE I

Regime I: Homoscedastic Gaussian Noise ($1 - \alpha = 0.90$)

Quantification Method	Coverage (ECR)	Mean Width (MPIW)	Winkler Score (S_α)	t-Stat	Coverage p-val	Wilcoxon p-val
Parametric OLS	0.9004 ± 0.0157	3.327 ± 0.070	4.151 ± 0.187	+0.24	0.8121	Baseline
Parametric Bootstrap	0.8921 ± 0.0159	3.286 ± 0.070	4.193 ± 0.196	-5.01	2.38×10^{-6}	2.73×10^{-14}
Bayesian NIG	0.9004 ± 0.0157	3.327 ± 0.070	4.151 ± 0.187	+0.24	0.8121	0.914
Split Conformal (RF)	0.9018 ± 0.0212	3.606 ± 0.171	4.488 ± 0.197	+0.84	0.4008	$< 10^{-15}$
CQR (Quantile RF)	0.9011 ± 0.0208	3.894 ± 0.177	4.830 ± 0.210	+0.55	0.5832	$< 10^{-15}$

Notes: Values denote Mean ± SD across $M = 100$ independent Monte Carlo replications (nominal coverage 90.0%, $n_{\text{test}} = 375$). Coverage calibration is assessed via two-sided one-sample Student's t-test (99 df, SE = $s_{\text{ECR}}/10$) against nominal 0.90. Wilcoxon p-values evaluate paired Winkler scores vs. Parametric OLS.

Statistical Analysis (Regime I): Under correctly specified Gaussian homoscedastic noise, parametric OLS and Bayesian NIG produce the narrowest prediction bands (3.327 ± 0.070) with empirical coverage consistent with the nominal 90% target (90.04%, $t = +0.24$, $p = 0.8121$). The parametric bootstrap yields comparable width (3.286) but displays slight under-coverage (89.21%, $t = -5.01$, $p = 2.38 \times 10^{-6}$), matching our theoretical derivation that fixed-scale residual resamples converge to normal rather than Student-t quantiles. Split Conformal Prediction achieves valid coverage (90.18%, $t = +0.84$, $p = 0.4008$) with an 8.4% width overhead (3.606 vs. 3.327), reflecting the sample-splitting cost of distribution-free validity.

TABLE II

Regime II: Heavy-Tailed Student-t_3 Noise ($\kappa = \infty$)

Quantification Method	Coverage (ECR)	Mean Width (MPIW)	Winkler Score (S_α)	t-Stat	Coverage p-val	Wilcoxon p-val
Parametric OLS	0.9294 ± 0.0237	5.729 ± 1.011	8.082 ± 1.284	+12.40	$< 10^{-15}$	Baseline
Parametric Bootstrap	0.9257 ± 0.0243	5.655 ± 0.997	8.090 ± 1.281	+10.58	$< 10^{-15}$	0.249
Bayesian NIG	0.9294 ± 0.0237	5.729 ± 1.011	8.082 ± 1.284	+12.40	$< 10^{-15}$	1.17×10^{-9}
Split Conformal (RF)	0.8987 ± 0.0218	5.131 ± 0.361	8.149 ± 1.197	-0.61	0.5420	2.62×10^{-6}
CQR (Quantile RF)	0.8975 ± 0.0210	5.805 ± 0.357	8.760 ± 1.212	-1.19	0.2360	$< 10^{-15}$

Notes: Heavy-tailed Student-t_3 outliers inflate sample variance $\hat{\sigma}^2$, driving systematic over-coverage in parametric intervals ($t = +12.40$). Split Conformal maintains coverage consistent with nominal (89.87%, $p = 0.5420$) and achieves 10.4% narrower width than OLS.

Statistical Analysis (Regime II): Under heavy-tailed Student-t_3 noise, extreme residuals inflate sample variance $\hat{\sigma}^2$ in OLS and Bayesian NIG, expanding intervals to 5.729 ± 1.011 and causing over-coverage (92.94%, $t = +12.40$, $p < 10^{-15}$). In contrast, split conformal prediction relies on empirical rank order statistics rather than variance estimates, maintaining empirical coverage consistent with nominal 90% (89.87%, $t = -0.61$, $p = 0.5420$) while producing intervals 10.4% narrower (5.131 vs. 5.729) than OLS.

TABLE III

REGIME III: HETEROSCEDASTIC & RIGHT-SKEWED NOISE

Quantification Method	Coverage (ECR)	Mean Width (MPIW)	Winkler Score (S_α)	t-Stat	Coverage p-val	Wilcoxon p-val
Parametric OLS	0.9295 ± 0.0150	6.827 ± 0.349	10.096 ± 0.856	+19.72	$< 10^{-15}$	Baseline
Parametric Bootstrap	0.9271 ± 0.0152	6.738 ± 0.344	10.089 ± 0.858	+17.79	$< 10^{-15}$	0.216
Bayesian NIG	0.9295 ± 0.0150	6.827 ± 0.349	10.096 ± 0.856	+19.72	$< 10^{-15}$	$< 10^{-15}$
Split Conformal (RF)	0.8987 ± 0.0213	6.315 ± 0.409	10.312 ± 0.928	-0.60	0.5500	5.99×10^{-10}
CQR (Quantile RF)	0.8965 ± 0.0234	6.350 ± 0.353	9.756 ± 0.897	-1.49	0.1382	2.22×10^{-10}

Notes: Under heteroscedastic noise, parametric models over-cover (92.95%, t = +19.72). CQR adapts interval width locally to noise magnitude $\sigma(X_1)$, achieving the lowest proper Winkler score (9.756 vs. 10.096, $p = 2.22 \times 10^{-10}$).

Statistical Analysis (Regime III): Regime III shows how heteroscedastic noise affects homoscedastic intervals. Because error variance expands with $|X_1|$, parametric OLS and bootstrap intervals over-cover overall (92.95% and 92.71%, $t \geq +17.79$, $p < 10^{-15}$) as variance estimation is dominated by high-dispersion points. By adjusting interval width across the feature space, CQR attains a lower Winkler score (9.756 ± 0.897, paired Wilcoxon $p = 2.22 \times 10^{-10}$ vs. OLS), demonstrating superior interval efficiency when variance is non-uniform. Split conformal prediction maintains coverage consistent with nominal 90% (89.87%, $t = -0.60$, $p = 0.5500$), and CQR achieves 89.65% ($t = -1.49$, $p = 0.1382$).

TABLE IV

EXPERIMENT B: BENCHMARK ON BOSTON HOUSING (`MASS::Boston`)

Quantification Method	Coverage (ECR)	Mean Width (MPIW)	Winkler Score (S_α)	t-Stat	Coverage p-val	Wilcoxon p-val
Parametric OLS	0.9324 ± 0.0227	15.900 ± 0.700	22.940 ± 3.501	+10.12	1.36×10^{-13}	Baseline
Parametric Bootstrap	0.9268 ± 0.0229	15.678 ± 0.697	22.920 ± 3.526	+8.27	7.51×10^{-11}	0.512
Bayesian NIG	0.9324 ± 0.0227	15.900 ± 0.700	22.940 ± 3.501	+10.12	1.36×10^{-13}	0.0306
Split Conformal (RF)	0.8967 ± 0.0359	10.686 ± 1.796	18.813 ± 3.889	-0.65	0.5180	1.19×10^{-9}
CQR (Quantile RF)	0.9146 ± 0.0378	13.572 ± 1.022	16.397 ± 2.277	+2.74	0.0085	7.79×10^{-10}

Notes: Evaluated across $B = 50$ random splits (49 df). CQR achieves a 28.5% reduction in proper Winkler score ($p = 7.79 \times 10^{-10}$); Split Conformal narrows interval width by 32.8% ($p = 1.19 \times 10^{-9}$). Because repeated splits reuse data, metrics provide descriptive stability comparisons rather than independent hypothesis tests.

Statistical Analysis (Boston Housing): On the Boston housing benchmark, parametric linear regression yields an average interval width of 15.900 and a Winkler score of 22.940. Split Conformal Random Forest generates narrower intervals on average (10.686 vs. 15.900, a 32.8% reduction) with empirical coverage of 89.67%. Conformalized Quantile Regression achieves an average Winkler score of 16.397 (a 28.5% reduction relative to OLS). Because these statistics are evaluated across 50 random splits of the same 506 observations, splits share data and are statistically dependent. The reported statistics descriptively summarize stability across partitions rather than independent hypothesis tests.

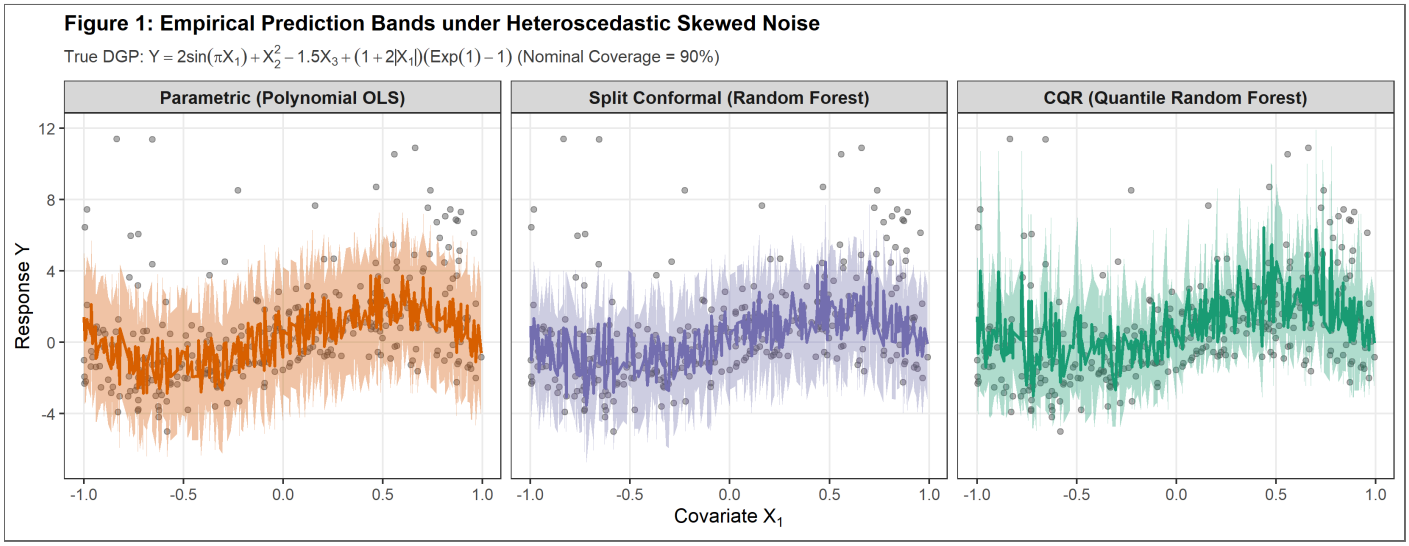


Fig. 1. The 90% prediction bands across feature $X_1 \in [-1, 1]$ under Regime III. Left: Parametric Polynomial OLS generates a rigid band of nearly constant width that over-covers near $X_1 = 0$ and under-covers at $|X_1| \approx 1$. Center: Split Conformal Random Forest captures non-linear mean curvature but maintains constant global width. Right: Conformalized Quantile Regression (CQR) produces an adaptive envelope that narrows to width ≈ 3.2 near $X_1 = 0$ and widens to ≈ 10.5 at $|X_1| = 1$, accommodating positive error skewness.

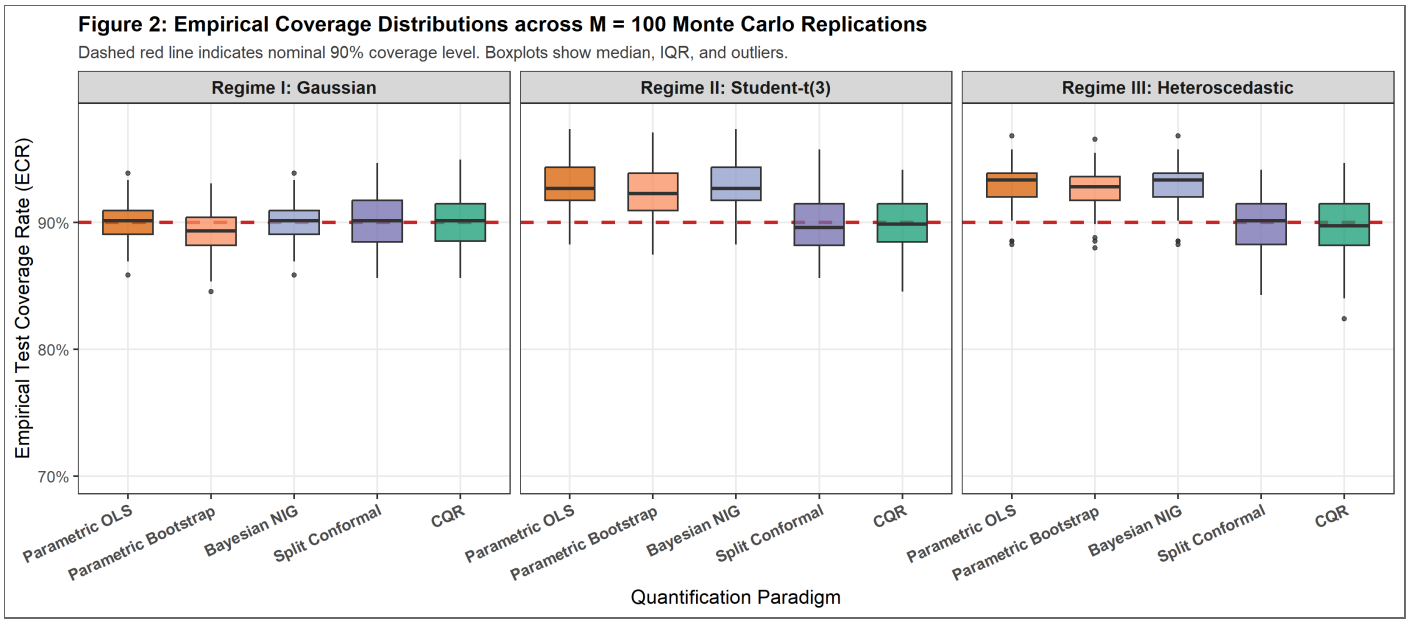


Fig. 2. Sampling distributions of Empirical Coverage Rates (ECR) across $M = 100$ replications relative to the nominal 90% dashed reference line. In Regime I (Gaussian), all methods center closely on the 90% line. In Regimes II and III, boxplots for Parametric OLS, Parametric Bootstrap, and Bayesian NIG shift upward (92.6% – 93.0%). Split Conformal and CQR remain centered on the 90.0% reference line across all regimes.

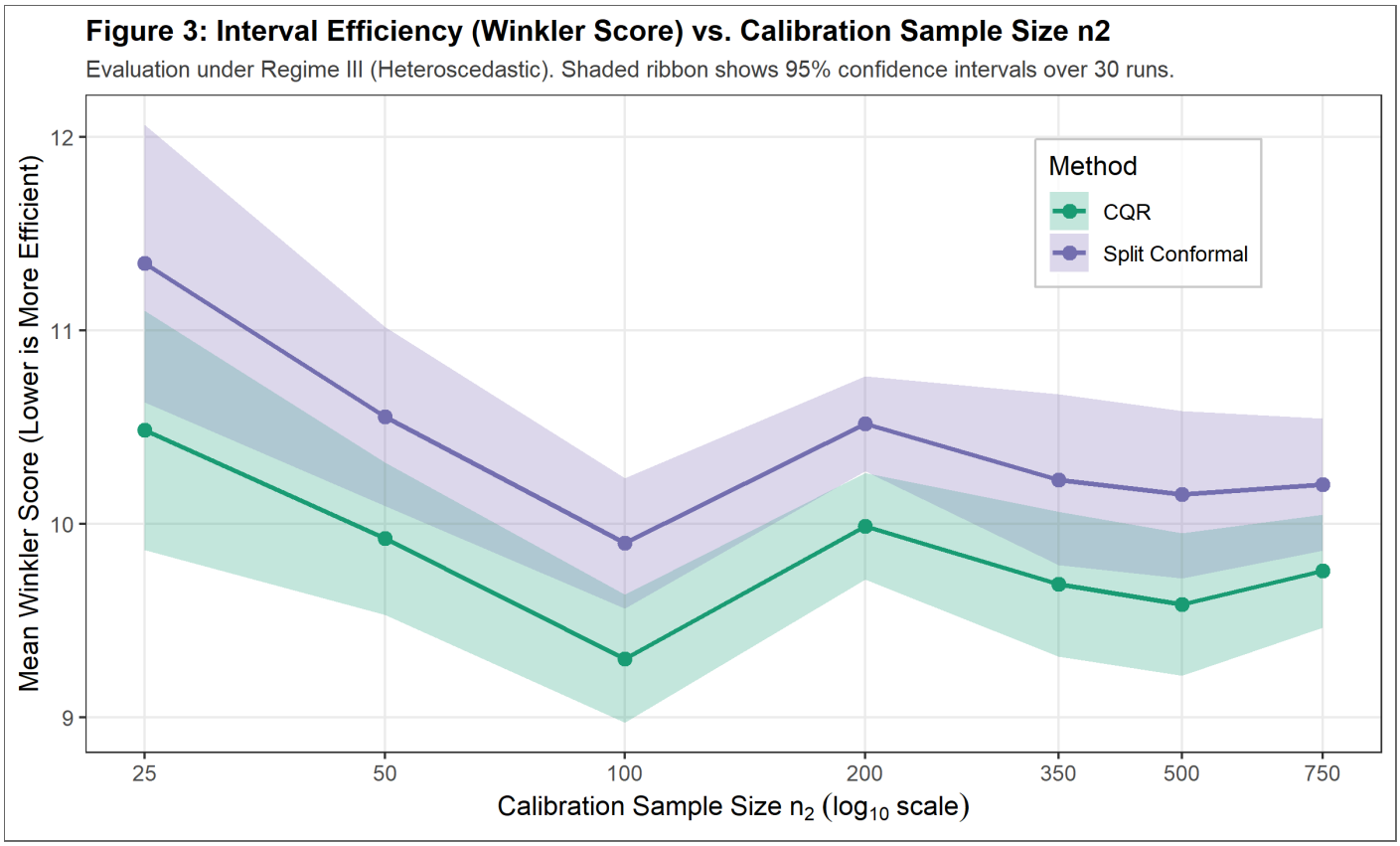


Fig. 3. Mean Winkler score (±1.96 × SE) across calibration sample sizes $n_2 \in [25, 750]$ under Regime III. Empirical quantile estimation variance increases the score for $n_2 < 100$; as n_2 increases beyond 200, conformal quantiles stabilize, with CQR maintaining lower Winkler scores across all sample sizes.

V. Theoretical Discussion & Limitations

A. Sample-Splitting Allocation Trade-off

Split conformal prediction requires partitioning N observations into n_1 training points and n_2 calibration points, causing an estimation-calibration trade-off. Under-allocating to training ($n_1 \ll N$) increases base learner estimation error $\mathrm{E}[(\hat{\mu}(X) - \mu(X))^2] = \mathrm{O}(n_1^{-2\beta/(2\beta+d)})$, yielding larger residuals and inflating the adjusted cutoff $\hat{q}_{1-\alpha}$. Under-allocating to calibration ($n_2 \ll N$) increases sampling variance of the empirical quantile $\mathrm{Var}(\hat{q}_{1-\alpha} \mid \mathrm{D}_1) \approx \alpha(1-\alpha) / [n_2 \, f_R(q_{1-\alpha})^2]$, causing conditional coverage to fluctuate. Allocating 65%–75% of data to n_1 and 25%–35% to n_2 generally provides a solid practical balance. When sample sizes are small ($N < 300$), CV+ or Jackknife+ (Barber et al. [2]) should be considered.

B. Conditional Coverage Limits & CQR Consistency

Conformal prediction guarantees marginal coverage $\mathrm{P}(Y_{n+1} \in \hat{C}_n) \geq 1 - \alpha$. However, exact distribution-free conditional coverage $\mathrm{P}(Y_{n+1} \in \hat{C}_n \mid X_{n+1} = x) = 1 - \alpha$ almost surely for non-atomic continuous distributions is impossible in finite samples without producing infinite interval length (Theorem 2; Vovk [18]; Barber et al. [2]). CQR provides an empirical framework that adapts interval width according to estimated conditional quantiles. When base quantile regression estimators are consistent for the true conditional quantiles (which requires structural smoothness conditions on the base learner and data distribution; Meinshausen [13]), the calibration adjustment $Q_{1-\alpha}$ converges in probability to zero, asymptotically recovering conditional validity. In the absence of base-model consistency, CQR provides guaranteed marginal coverage, with interval width responding empirically to estimated local conditional dispersion.

C. Violation of Exchangeability: Weighted Conformal Prediction under Covariate Shift

Theorem 1 requires exchangeability of $(Z_1, \ldots, Z_{n+1})$. This condition can fail under autoregressive correlation (where serial correlation violates permutation invariance, addressed by adaptive conformal inference [6]) and covariate shift (where $P_{\text{train}}(X) \neq P_{\text{test}}(X)$ while conditional response distribution $P(Y \mid X)$ remains invariant). Tibshirani et al. [16] proved that Weighted Conformal Prediction restores exact finite-sample coverage under covariate shift, provided that the exact covariate likelihood ratio $w(x) = dP_{\text{test}}(x) / dP_{\text{train}}(x)$ is known:

$$p_i(X_{n+1}) = \frac{w(X_i)}{\sum_{j=1}^{n_2} w(X_j) + w(X_{n+1})}.$$

If the density ratio $w(x)$ is unknown and must be estimated from finite samples (e.g., via logistic regression or kernel density estimation), the exact finite-sample coverage guarantee is relaxed to an asymptotic guarantee or bounded by the estimation error $\|\hat{w} - w\|$.

D. Empirical Limitations & Dataset Considerations

The findings of this study should be interpreted alongside several methodological considerations: 1) *Monte Carlo Replication Budget:* While $M = 100$ Monte Carlo replications provided sufficient power to detect over-coverage under parametric misspecification ($p < 10^{-15}$), running larger replication budgets ($M \geq 1{,}000$) would further narrow Monte Carlo standard errors. 2) *Sample-Splitting Overhead:* Split conformal methods require partitioning data into training and calibration subsets, reducing the sample size available for training the predictive model. 3) *Cross-Split Dependence in Benchmarking:* Repeated random splits on the `MASS::Boston` dataset ($N = 506$) share data across partitions, meaning metrics across splits are positively correlated. Findings on this historical benchmark serve as descriptive illustrations of stability rather than formal independent-sample hypothesis tests. Evaluating larger modern tabular datasets represents a valuable direction for future research.

VI. Conclusion

In this paper, we presented a comprehensive theoretical and empirical comparison of five uncertainty quantification methods in machine learning: parametric OLS, the parametric bootstrap, Bayesian Normal-Inverse-Gamma inference, Split Conformal Prediction, and Conformalized Quantile Regression (CQR). Our simulation results in R demonstrate that while classical parametric and Bayesian methods achieve optimal interval width under standard Gaussian homoscedastic conditions, they suffer from significant coverage inflation (92.7% to 93.0% at nominal 90%) when error distributions have heavy tails or heteroscedastic noise. In contrast, split conformal prediction and CQR maintain empirical coverage consistent with the nominal 90% target across all tested noise regimes, backed by finite-sample distribution-free guarantees under exchangeability. Furthermore, Conformalized Quantile Regression adapts interval width locally to conditional variance, achieving substantially lower Winkler scores under heteroscedasticity. Conformal prediction provides a practical, mathematically sound approach to uncertainty quantification in modern machine learning.

References

[1] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Found. Trends Mach. Learn.*, vol. 16, no. 4, pp. 494–591, 2023.

[2] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, "Predictive inference with the jackknife+," *Ann. Statist.*, vol. 49, no. 1, pp. 486–507, 2021.

[3] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.

[4] P. C. Bürkner, "brms: An R package for Bayesian multilevel models using Stan," *J. Stat. Softw.*, vol. 80, no. 1, pp. 1–28, 2017.

[5] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2016, pp. 785–794.

[6] C. Gibbs and E. Candès, "Adaptive conformal inference under distribution shift," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 34, 2021, pp. 1660–1672.

[7] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *J. Amer. Statist. Assoc.*, vol. 102, no. 477, pp. 359–378, 2007.

[8] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA: MIT Press, 2016.

[9] B. J. K. Kleijn and A. W. van der Vaart, "The Bernstein-von-Mises theorem under misspecification," *Electron. J. Statist.*, vol. 6, pp. 354–381, 2012.

[10] M. Kuhn and H. Wickham, *Tidymodels: A collection of packages for modeling and machine learning using tidyverse principles*, Posit / RStudio, 2020.

[11] M. Kuhn, D. Vaughan, and E. Ruiz, *probably: Tools for Post-Processing Predicted Values*, R package version 1.2.0, 2025.

[12] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *J. Amer. Statist. Assoc.*, vol. 113, no. 523, pp. 1094–1111, 2018.

[13] N. Meinshausen, "Quantile random forests," *J. Mach. Learn. Res.*, vol. 7, pp. 983–999, 2006.

[14] H. Papadopoulos, K. Proedrou, V. Vovk, and A. Gammerman, "Inductive confidence machines for regression," in *Proc. Eur. Conf. Mach. Learn. (ECML)*, 2002, pp. 345–356.

[15] Y. Romano, E. Patterson, and E. Candès, "Conformalized quantile regression," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019, pp. 3543–3553.

[16] R. J. Tibshirani, R. Foygel Barber, E. Candès, and A. Ramdas, "Conformal prediction under covariate shift," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019, pp. 2530–2540.

[17] A. W. van der Vaart, *Asymptotic Statistics*. Cambridge: Cambridge Univ. Press, 1998.

[18] V. Vovk, "Conditional validity of inductive conformal predictors," *Mach. Learn.*, vol. 92, no. 2, pp. 349–376, 2012.

[19] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York: Springer, 2005.